

Dataset-Routed Clustering with Candidate Pair Constraints for AnimalCLEF 2026

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Abstract

AnimalCLEF 2026 evaluates individual animal discovery and re-identification as an open-set clustering task across four datasets: **LynxID2025**, **SalamanderID2025**, **SeaTurtleID2022**, and **TexasHornedLizards**. Each test image must be assigned to an identity cluster, and submissions are evaluated using Adjusted Rand Index. We describe a dataset-routed clustering pipeline designed for the different visual cues and data conditions of the four species. The pipeline combines pretrained wildlife descriptors, supervised metric-learning embeddings, local ORB matching, XGBoost pair scoring, self-training, and dataset-specific post-processing. Lynx and sea turtle were handled mainly with descriptor-based clustering, while salamander and Texas horned lizard required additional local pair evidence and graph constraints. For ambiguous candidate pairs, reviewed must-link and cannot-link decisions were compiled into the final clustering graph. The final submission by team VIPL-VSU achieved 0.67987 public ARI and 0.69632 private ARI, ranking third on the private leaderboard.

Keywords

animal re-identification, open-set clustering, metric learning

1. Introduction

Individual animal re-identification aims to recognise the same animal across photographs collected at different times, locations, viewpoints, and acquisition conditions. It is an important tool for ecological monitoring because many species can be identified from persistent visual markings such as coat patterns, scales, scutes, stripes, spots, or ventral markings. Compared with human re-identification, animal re-identification is often more open-ended: the number of individuals is not fixed, many individuals have only a few images, backgrounds and camera traps can be highly biased, and the visual cue differs substantially across species [1, 2].

AnimalCLEF 2026, organised within LifeCLEF 2026, casts this general problem as discovery-oriented clustering across four animal datasets [3, 4]. This setting makes the objective different from ordinary retrieval: a model with high nearest-neighbour recall can still fail if its clustering threshold merges different individuals, while a conservative pipeline can fail by over-splitting the same animal. The official metric is Adjusted Rand Index (ARI), a chance-corrected measure of agreement between two partitions [5].

Our final pipeline was built around one observation: the four datasets expose different visual and statistical structures, so a single backbone and threshold is not robust. Lynx images contain relatively stable coat patterns and strong cluster-count priors. Salamanders require fine local texture and careful ambiguity handling. Sea turtles are already well served by frozen wildlife descriptors. Texas horned lizards have no labelled local train split in the released package, making them a strong domain-shift and self-training problem. We therefore use a dataset-routed pipeline in which each species branch has its own representation, pair evidence, thresholding, and post-processing policy.

The key simplification is to avoid treating clustering as one monolithic thresholding problem. We instead split it into a high-recall first stage and a precision-oriented second stage. The first

stage is allowed to be inclusive: global descriptors, local ORB evidence, and species-specific preprocessing only need to surface plausible same-individual links. The second stage is selective: candidate pair verification converts a much smaller ambiguity set into positive and negative graph constraints.

The verification component of our pipeline was inspired by ambiguity-aware pair sampling for animal re-identification [6]. That work frames feedback as pairwise constraints, producing must-link and cannot-link signals for constrained clustering. We adopt the same pairwise interface in a narrower engineering form: the model proposes candidate pairs, and the verification stage only accepts or rejects pair edges. In our competition setting, these decisions are compiled into clustering constraints over top- K local matches and uncertain clusters.

Our contributions are:

- A practical dataset-routed clustering pipeline for AnimalCLEF 2026, with official scores of 0.67987 public ARI and 0.69632 private ARI.
- A two-stage clustering decomposition that uses automatic top- K candidate retrieval for recall and pair-constraint verification for precision.
- An account of the species-specific routes that proved useful: transductive seed smoothing for lynx, learned pairwise fusion and graph constraints for salamanders, conservative descriptor fusion for sea turtles, and trusted-view self-training for Texas horned lizards.
- Documented intervention evidence showing that many useful components were not universal: ORB local matching helped salamanders but was not used for lynx or sea turtles, yellow-pattern cues hurt as hard vetoes but helped as learned XGBoost features, and Texas descriptor concatenation degraded despite preserving seed pairs.

2. Related Work

2.1. Animal Re-Identification

Animal re-identification has historically relied on species-specific visual cues and semi-automatic photo-identification tools, especially for animals with stable markings [1]. Recent benchmarks and toolkits have made the setting more comparable across species. WildlifeDatasets collects many individual animal datasets and provides a common evaluation interface, and MegaDescriptor is one of the strongest general descriptors from this line of work [2]. MiewID is another multispecies model trained from community-curated identity data and was complementary to MegaDescriptor in our experiments [7]. Dataset-specific benchmarks are also important: SeaTurtleID2022 emphasizes long-span open-set re-identification under realistic temporal splits [8], while CzechLynx provides a large camera-trap resource for Eurasian lynx individual identification and related vision tasks [9].

2.2. Local Matching and Species-Specific Cues

Local feature matching is useful when identity is encoded in spatially localized markings. ORB provides a fast binary keypoint descriptor that can be paired with geometric verification [10], while modern matchers such as LightGlue offer stronger learned correspondences for image matching [11]. For Texas horned lizards, the official reference study by Biffi et al. showed that ventral spot patterns carry individual identity information: HotSpotter placed the correct previous image first for 85 of 90 known recapture photos, and in four additional cases the correct match was still among the top-10 recommendations [12]. This supports a useful separation between local matching as candidate retrieval and later decisions about which candidate links are reliable.

2.3. Constrained Clustering and Pair Feedback

Open-set re-identification pipelines often need to convert retrieval evidence into a partition, so pair-level decisions become useful when descriptor scores alone are ambiguous. Ambiguity-aware pair sampling for animal re-identification converts selected pairwise feedback into must-link and cannot-link constraints for clustering [6]. We use this line of work as the conceptual basis for treating verified candidate pairs as graph constraints.

3. Task and Data

3.1. Task Definition

The goal of AnimalCLEF 2026 is to discover individual animals in the hidden test split and submit the resulting identity partition. In other words, participants are not asked to predict species labels or to assign images to a fixed set of known training identities. They must group the unlabelled test images so that photographs of the same animal receive the same cluster identifier and photographs of different animals receive different identifiers.

The input consists of four dataset namespaces, labelled training identities where available, and unlabelled test images. The required submission is a CSV file with two columns, `image_id` and `cluster`. The official evaluation compares the submitted partition with the hidden ground-truth identity partition using Adjusted Rand Index (ARI). We therefore treat AnimalCLEF 2026 as an open-set clustering task, where retrieval scores are only intermediate evidence and the final object being evaluated is the complete test-set clustering.

3.2. Dataset Splits

The officially released data contain 15,483 images in four dataset namespaces. The train split contains 13,074 labelled images for three datasets, while `TexasHornedLizards` is provided as a test-only dataset with 274 images. The test split contains 2,409 images in total. The datasets are released through the AnimalCLEF benchmark [4]; where available, we also cite the underlying dataset or species study used by the benchmark [13, 8, 9, 12].

Table 1

Dataset sizes in the released AnimalCLEF 2026 package used by our pipeline.

Dataset	Train images	Test images	Main visual cue / source
<code>LynxID2025</code>	2,957	946	coat/flank markings; CzechLynx-related [9]
<code>SalamanderID2025</code>	1,388	689	dorsal yellow spot patterns [13]
<code>SeaTurtleID2022</code>	8,729	500	facial and scute patterns [8]
<code>TexasHornedLizards</code>	0	274	ventral/body spot patterns [12]
Total	13,074	2,409	–

The labelled train distribution is strongly long-tailed. Across the three labelled datasets there are 1,102 identities and 317 singleton identities. Salamander is the most extreme case: `SalamanderID2025` has 587 train identities, of which about 52.8% are singletons. This matters because identity-level validation and pairwise training produce very different support across datasets. Figure 1 shows the per-dataset identity-size distribution that motivated the dataset-routed design.

Local validation protocol. Local validation refers to our internal experiment protocol, not the official Kaggle evaluation split. For each labelled dataset, identities are shuffled with seed 42 and approximately 10% of identities are held out; all images of the same individual stay on the same side of the split. This produces 991 training-side identities with 11,739 training-side images

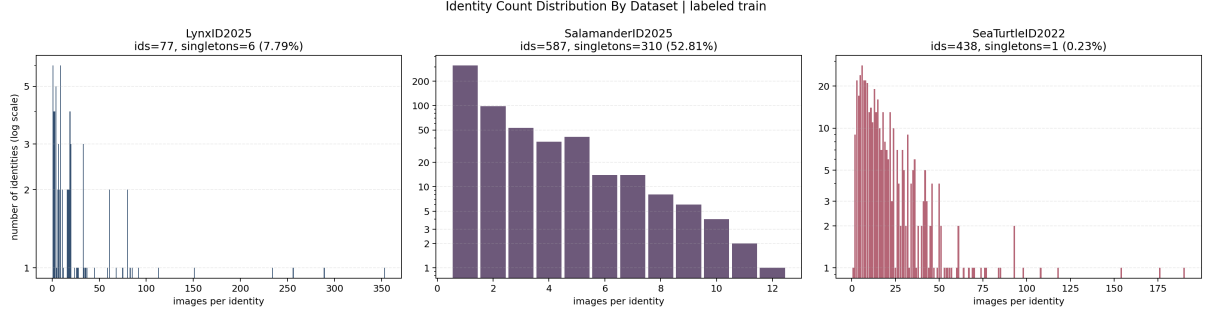


Figure 1: Identity-size distribution in the labelled training split. Salamander is dominated by singleton identities, while sea turtle has many multi-image identities and lynx has fewer identities with a long-tailed image count. This motivates identity-level validation and species-specific clustering policies.

and 111 validation identities with 1,335 validation images. The validation split contains 69/8 training/validation identities for lynx, 528/59 for salamander, and 394/44 for sea turtle. Official competition scores are reported separately from the Kaggle public and private leaderboards.

We report ARI as the main clustering metric. We also track Normalized Mutual Information, pairwise precision/recall/F1, Recall@1/5/10, cluster count, and singleton-cluster ratio. Retrieval metrics are treated only as candidate-recall diagnostics: a high Recall@10 means that the global route can supply useful candidates, but the final objective remains the ARI of the cluster partition.

4. Method

The final system is a dataset-routed clustering pipeline. Each image is first assigned to one of the four dataset namespaces, and each dataset is then processed with the route that worked best for its visual cue and data setting. Figure 2 gives the full flow, and Table 2 lists the submitted route for each dataset.

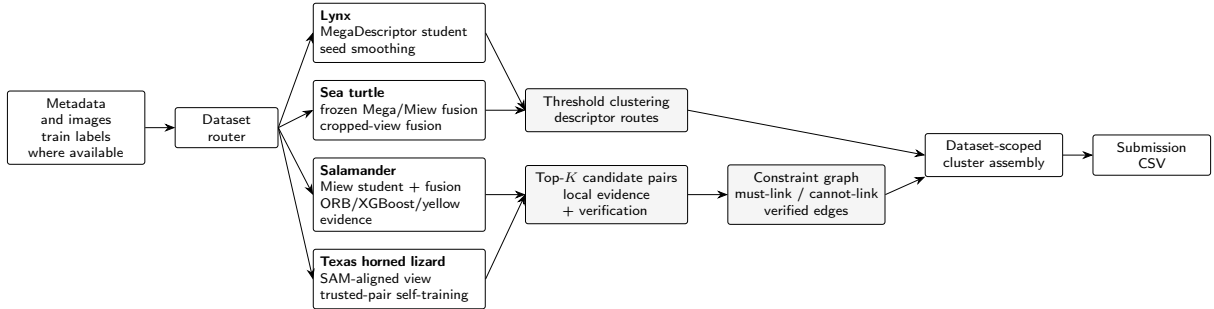


Figure 2: Pipeline overview. Images are first routed by dataset. Lynx and sea turtle use descriptor-based threshold clustering, while salamander and Texas horned lizard use high-recall top- K candidate pairs followed by pair verification and constraint-graph compilation. The dataset-scoped cluster outputs are then assembled into the submitted CSV.

Most routes start from MiewID-msv3 [7] and MegaDescriptor-L [2]. Embeddings are L2-normalised, compared by cosine similarity, and clustered independently for each dataset using average-linkage clustering [14]. Frozen descriptor fusion was already strong for sea turtle, while supervised metric-learning students were useful for lynx and salamander. The supervised routes use dataset-specific metric-learning heads and a 512-dimensional L2-normalised embedding; implementation details are given in Appendix A.

Local pair evidence and graph constraints. Descriptor thresholds alone were not enough for salamander and Texas horned lizard. For these two datasets, we retrieved top- K candidate

Table 2

Final submission recipe by dataset.

Dataset	Main blocks	Dim.	Threshold	Constraint or post-processing policy
LynxID2025	MegaDescriptor-L supervised student; transductive seed smoothing	512	0.85	average-linkage thresholding
SalamanderID2025	MiewID student; frozen descriptor fusion; ORB and yellow-pattern features; XGBoost pair scoring	4200	0.25	top- K pair graph with reviewed must-link/cannot-link constraints
SeaTurtleID2022	frozen MegaDescriptor+MiewID fusion with cropped-view fusion	7376	0.70	conservative average-linkage thresholding; no pair constraints
TexasHornedLizards	SAM-aligned gray center-body views; trusted-view MiewID self-training	512	0.44	thresholding plus reviewed pair-registry repair

Table 3

Identity-level validation ARI for frozen descriptor baselines. Texas is excluded because no labelled train split is available locally.

Route	Lynx ARI	Salamander ARI	SeaTurtle ARI
Frozen MiewID	0.0864	0.1681	0.7854
Frozen MegaDescriptor	0.1027	0.1078	0.8903
Frozen MegaDescriptor + MiewID	0.2250	0.2030	0.9256

neighbours and added local ORB evidence with RANSAC verification [10, 15]. For salamander, the final pair scorer was an XGBoost model [16] using descriptor similarities, route-specific scores, ORB evidence, and yellow-pattern features from masked views [17]. For Texas horned lizard, the useful route first built SAM-aligned gray center-body views and then trained a MiewID student from trusted candidate pairs and single-view consistency.

Candidate-pair decisions are used only as graph constraints. Positive decisions become must-link edges, negative decisions become cannot-link constraints, and the graph compiler merges only compatible components. The final constrained submission is not a fully automatic rerun: it includes human-in-the-loop review of retrieved candidate pairs for salamander and Texas horned lizard. The reviewer saw only candidate image pairs from the released competition data and marked them as same individual, different individual, or uncertain. Hidden labels were not available, and the reviewer did not directly edit final cluster identifiers. In the final submission, the salamander constraints contained 179 positive pairs and 4216 negative or default-negative pairs; the Texas registry contained 86 must-link and 1514 cannot-link rows. The compact constraint artifacts and the graph-compilation code are included in the public release.

Table 4

Submission trace by intervention stage. Scores are ARI, and each delta is relative to the previous row.

Stage	Public	Δ Pub.	Private	Δ Priv.	Main change
Data-routed base	0.37277	–	–	–	Dataset-specific descriptor routes and threshold clustering.
Texas self-training	0.45751	+0.08474	–	–	Add SAM-aligned Texas views, trusted top- K pseudo-pairs, and a Texas-specific MiewID student.
Salamander XGBoost fusion	0.48524	+0.02773	–	–	Add learned pair scoring over descriptor, route-difference, and ORB local-match features.
Yellow-ROI local evidence	0.51595	+0.03071	0.44839	–	Add colour-based yellow-pattern features to the salamander pair scorer.
Texas pair registry	0.62528	+0.10933	0.62091	+0.17252	Add reviewed Texas pair constraints for small graph repair.
Pre-final routed base	0.62817	+0.00289	0.65096	+0.03005	Last routed base before adding the final salamander pair constraints.
Salamander pair constraints	0.67578	+0.04761	0.69632	+0.04536	Add reviewed top- K salamander candidate-pair constraints.
Final submission	0.67987	+0.00409	0.69632	+0.00000	Add six extra salamander cross-view positive pairs; VIPL-VSU private rank 3.

Algorithm 1. Constrained pair-graph compilation

Input	base partition C_0 ; scored candidate pairs $P = \{(i, j, p_{ij})\}$; reviewed must-links M ; reviewed cannot-links N ; graph threshold τ .
Output	final partition C .
1 Initialise a disjoint-set U from the base clusters in C_0 . 2 Build a forbidden-neighbour map F from all cannot-link pairs in N . 3 Create ordered edge list E : all must-links in M first, then all scored pairs in P with $p_{ij} \geq \tau$, sorted by descending score. 4 for each edge (i, j) in E do 5 Let $A = \text{find}_U(i)$ and $B = \text{find}_U(j)$. If $A = B$, continue. 6 If A and B are not forbidden by F , merge them in U and update the forbidden-neighbour map. 7 Otherwise discard the edge as a conflict. 8 end for 9 Return the connected components induced by U , followed by deterministic split and attachment overlays.	

5. Results and Useful Interventions

Table 4 summarizes the main submission changes. It is an engineering trace rather than a controlled factorial ablation, so each delta should be read as the change from the previous submitted stage.

The useful interventions were dataset-specific. Texas improved most when the test-only images were converted into aligned body views and used for trusted self-training. Salamander improved through pair-level evidence: first XGBoost fused descriptor and ORB features, then yellow-pattern features made the pair scorer more informative. The largest final gain came from converting reviewed salamander candidate pairs into graph constraints. Sea turtle and lynx were more stable under descriptor-based clustering, so their useful changes were smaller.

Before manual pair review, the public-best routed base reached 0.51595 public ARI. Later route updates produced a pre-final base with public/private ARI 0.62817/0.65096. The final salamander pair constraints then raised the score to 0.67987/0.69632. This should be read as submission-level evidence, not as a controlled single-factor ablation. Our final submission was uploaded on 2026-04-29 03:04:38 UTC under team name VIPL-VSU and ranked third on the private leaderboard.

Additional cluster-shape diagnostics and negative probes are reported in Appendix C.

6. Discussion

The main observation is that AnimalCLEF 2026 is not only a representation-learning task. It is a partitioning problem under species-specific evidence, long-tailed identities, open-set test clusters, and incomplete local supervision. We found that strong pretrained descriptors were necessary but not sufficient. The most valuable work happened after candidate retrieval: deciding which pair evidence was trustworthy, which dataset should receive local matching, how to calibrate thresholds, and how to keep local validation evidence separate from official leaderboard observations.

Another lesson is that a test-only clustering graph can have an insufficient information budget for some individuals. A single weak view may not expose enough distinctive markings to decide a merge or split, so the useful route is often to expose better identity evidence before clustering. Sea turtle benefited from cropped-view fusion, Texas horned lizards from SAM-aligned body views, and salamanders from local yellow-pattern pair evidence.

The pair verification component should be interpreted narrowly. It was restricted to retrieved candidate pairs and ambiguous clusters, then compiled into deterministic graph operations. The decisions were pair-level constraints over visible image evidence, not hidden labels or direct identity assignments. This made the constraints inspectable, but it also means that the final constrained submission is human-in-the-loop. We therefore separate the automatic routed system from the final constrained submission in Table 4.

The largest limitation of this note is that it records a competition pipeline whose final branch selection depended on public leaderboard feedback and many local artifacts. We therefore report leaderboard gains mainly as intervention-level evidence rather than as a controlled factorial ablation. For reproducibility, this note reports the branch-level preprocessing, model choices, thresholds, pair-review counts, and graph-compilation settings used by the submitted pipeline. The final submitted CSV was uploaded on 2026-04-29 03:04:38 UTC and is associated with the public/private scores reported in Table 4.

7. Conclusion

We presented a dataset-routed hybrid clustering pipeline for AnimalCLEF 2026 that obtained 0.67987 public ARI and 0.69632 private ARI. The central lesson is that this benchmark was not solved by a single embedding or threshold. Performance improved when each dataset exposed its own identity evidence and when uncertain retrieval links were converted into graph constraints. For lynx and sea turtles, conservative descriptor clustering was sufficient; for salamanders and Texas horned lizards, local candidate-pair evidence and constraint compilation were the decisive parts of the final pipeline.

Code and Artifact Availability

We prepared a public release for the VIPL-VSU AnimalCLEF 2026 working note at <https://github.com/geyaokai/animalclef26>. The release includes the reusable pipeline code, runnable scripts, smoke tests, the final submitted CSV, compact salamander and Texas pair-constraint artifacts, route summaries, and selected checkpoints that fit GitHub LFS limits. Two checkpoints are tracked with Git LFS. The MegaDescriptor-based lynx checkpoint is larger than the GitHub LFS per-object limit, so its configuration and reports are included and the checkpoint should be distributed through a separate artifact host. Raw images, large caches, generated review boards, and private local staging/audit artifacts are not included.

Acknowledgments

We thank the AnimalCLEF 2026 organizers for preparing the benchmark, maintaining the Kaggle competition, and providing the evaluation infrastructure. We also thank the maintainers and authors of the open-source animal re-identification resources and pretrained models used in this work, including MiewID, MegaDescriptor, WildlifeDatasets, scikit-learn, OpenCV, and XGBoost.

Declaration on Generative AI

During the preparation of this work, the authors used generative AI tools for coding assistance, experiment organization, and language editing. The method design, experiments, result interpretation, and final manuscript content were reviewed and are the responsibility of the authors.

A. Implementation Details

Table 5

Main branch configuration for the final submitted pipeline.

Dataset	Route	Dim.	Threshold
LynxID2025	MegaDescriptor-L supervised student with seed smoothing	512	0.85
SalamanderID2025	Miew masked-SupCon student + frozen fusion + ORB/XGBoost/yellow features + graph constraints	4200	0.25
SeaTurtleID2022	frozen MegaDescriptor+MiewID fusion with cropped fusion branch, weight 0.3	7376	0.7
TexasHornedLizards	Miew trusted-view self-training on SAM-aligned gray center-body views	512	0.44

MegaDescriptor-L takes 384×384 RGB inputs and outputs a 1536-dimensional vector. MiewID-msv3 takes 440×440 RGB inputs and outputs a 2152-dimensional vector. Early fusion concatenates the two normalised blocks into a 3688-dimensional vector and normalises the result again. For supervised routes, each student backbone is followed by a linear projection to a 512-dimensional L2-normalised embedding. Training uses separate ArcFace heads [18] for each dataset rather than one shared closed-set head. We also evaluated feature distillation, relation distillation, masked supervised contrastive loss [19], and a salamander sub-center ArcFace head [18].

ORB extraction used up to 1024 features per image after resizing the longer side to 768 pixels. Candidate matching was implemented with OpenCV [20] using brute-force descriptor matching, a ratio test of 0.8, and RANSAC geometric verification [15]; pairs with fewer than 8 inliers received zero local support. The local score was used as a conservative boost:

$$s_{\text{rerank}} = s_{\text{global}} + w_{\text{local}} s_{\text{local}} (1 - s_{\text{global}}). \quad (1)$$

Here s_{global} is the descriptor cosine similarity, s_{local} is the normalized ORB/RANSAC local evidence, and w_{local} controls how strongly local matches can boost a pair; the factor $(1 - s_{\text{global}})$ keeps the boost small for pairs that are already globally similar. In the salamander smoke audit, positive pairs had mean local score 0.3366, while negative pairs had mean local score 0.0111.

For the gate2 salamander compiler, we used `graph_threshold=0.25`, `min_judged_pairs=2`, and `min_no_pairs=2`. The gate compiled 42 candidates, skipped 61 low-evidence candidates, changed 89 salamander images, and produced 60 overlay operations. Compared with a split-only base, it reduced salamander clusters from 427 to 412 and singletons from 260 to 229.

Table 6

Key operating parameters of the submitted pipeline. Thresholds are cosine-distance clustering thresholds unless noted otherwise.

Dataset	Main evidence	Gate	Pair-constraint role
LynxID2025	MegaDescriptor supervised student, TTA variants, transductive seed smoothing	0.85	not used; descriptor graph was sufficient
SalamanderID2025	Miew supervised student, frozen fusion, ORB top- $K = 10$, yellow-pattern features, XGBoost pair fusion	graph 0.25; ORB inliers ≥ 8	positive/negative constraints for ambiguous candidates
SeaTurtleID2022	frozen MegaDescriptor+MiewID fusion plus cropped-view fusion with weight 0.3	0.7	not used; conservative descriptor fusion was selected
TexasHornedLizards	SAM-aligned gray center-body views, Miew trusted-view self-training, ORB candidate evidence	0.44	automatic trusted pairs first; later pair review for ambiguous candidates

Table 7

Manual pair-constraint artifacts used in the final constrained submission.

Dataset / stage	Reviewed pairs	Changed images	Effect
Salamander pair constraints	179 yes / 4216 no or default no	200	top- K candidate-pair review; public/private ARI 0.62817/0.65096 $\rightarrow$ 0.67987/0.69632
Texas pair registry	86 yes / 1514 no	8	small graph repair on ambiguous Texas clusters

B. Additional Route Details

For lynx, we also tested route co-association over multiple supervised routes and thresholds. The useful variants combined synth-warmup and previous-best TTA routes through average, mutual- k NN, and complete-link style predictions, then kept pairs supported by a high co-association vote threshold. These probes were secondary to the final pipeline because the lynx branch was already reasonably stable under supervised metric learning and seed smoothing.

For salamanders, the final route was built in five stages: frozen MegaDescriptor+MiewID fusion; ORB local reranking only for salamanders; a MiewID-based supervised embedding trained with masked supervised contrastive loss and concatenated with frozen fusion features; XGBoost pairwise fusion over global, local, and route-specific features; and split or pair-constraint judgments compiled into a graph overlay. A yellow-only veto looked plausible because this dataset is dominated by yellow dorsal markings, but it over-fragmented clusters and reduced local validation ARI from 0.5875 to 0.2905 in one probe. The same cue, when introduced as an XGBoost feature, improved the public leaderboard from 0.49726 to 0.51082.

For Texas horned lizards, the trusted-view student used 46 trusted images and 228 untrusted images, producing 1142 training rows after view construction. The final selected route used trusted supervision plus all-image single-view pseudo-positive consistency, without teacher distillation. The branch was valuable not only for its own leaderboard gain but also because it established the top- K candidate retrieval and pair-verification interface later reused on salamanders.

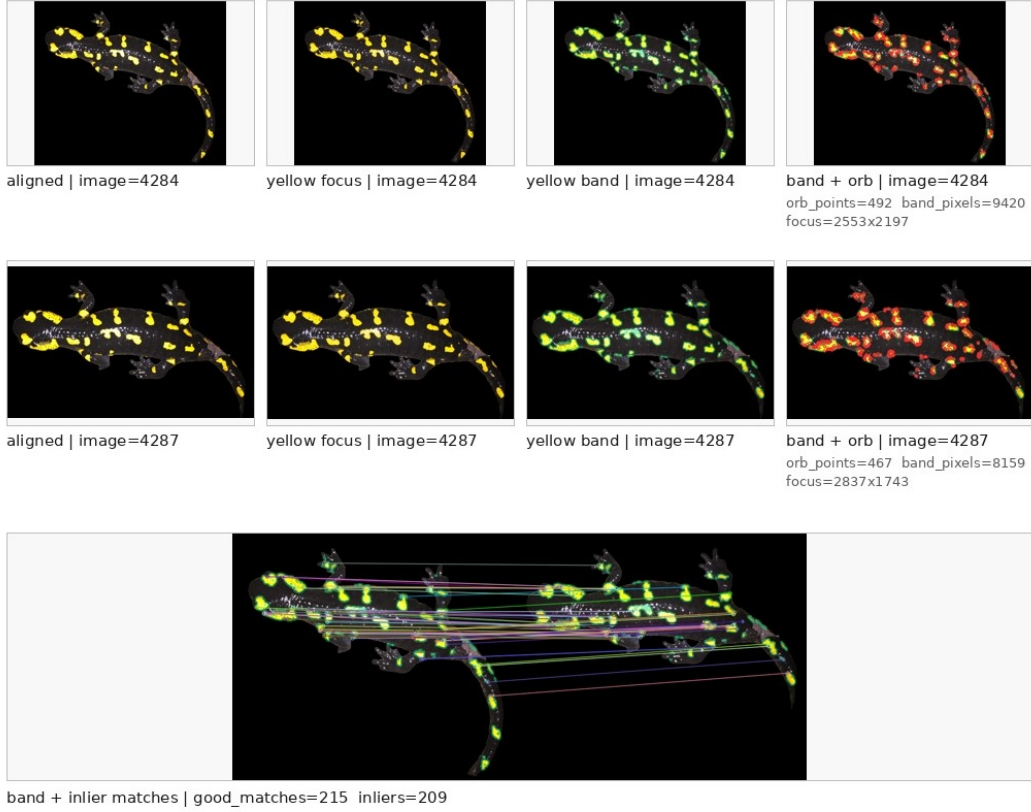


Figure 3: Appendix example of salamander pair evidence. Yellow-pattern views and ORB inlier matches provide local evidence for candidate pairs, but this cue was used as a pair feature rather than as a hard global veto.

C. Additional Results and Probes

Table 8

Cluster shape of the final constrained submission.

Dataset	Samples	Clusters	Singleton clusters	Singleton ratio
LynxID2025	946	21	2	0.095
SalamanderID2025	689	551	482	0.875
SeaTurtleID2022	500	153	41	0.268
TexasHornedLizards	274	213	175	0.822

Several negative results shaped the final pipeline. Descriptor concatenation for Texas preserved high-confidence seed pairs but degraded the public score, suggesting that the proxy did not penalise outsider merges near seed clusters. Salamander yellow cues were harmful as hard vetoes but helpful as learned pairwise features. Deep local matching probes with ALIKED [21] and LightGlue [11], plus a WildFusion-style calibration, did not outperform the simpler ORB-based route in our validation protocol. Dual-view SAM-masked salamander routes were engineering-ready but did not improve official submissions when directly inserted into the current XGBoost pipeline.

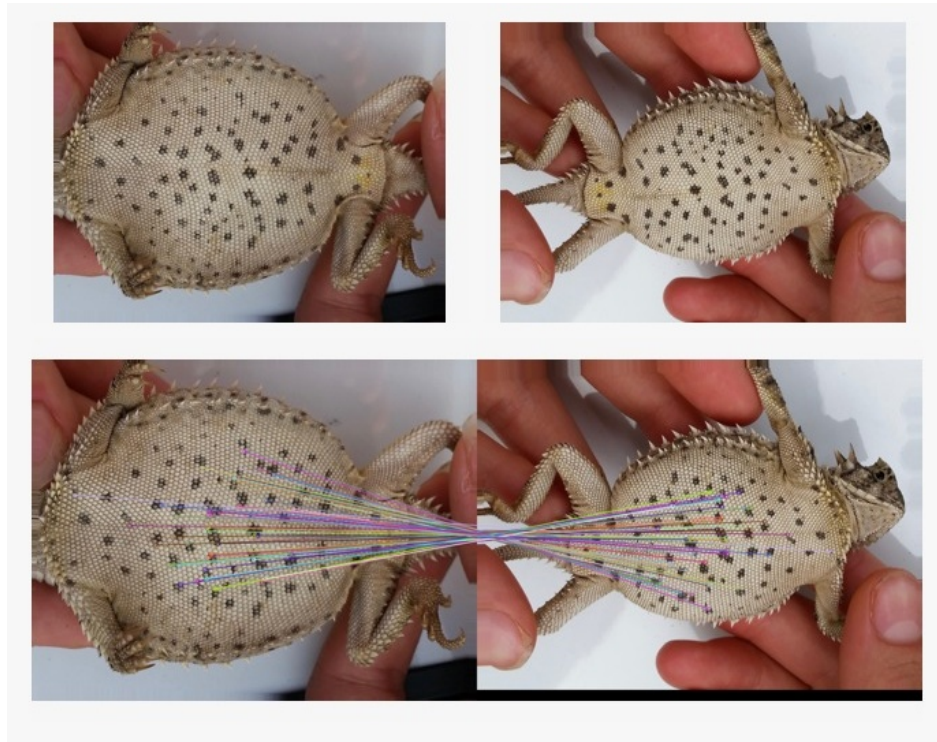


Figure 4: Appendix example of Texas horned lizard pair evidence. After view normalization, ORB inlier matches highlight the ventral-pattern evidence used for top- K candidate verification.

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